Report

Project: ULAB Blood Bank

Course Title: Smartphone Application Development
Course Code: CSE 438
Section: 01

Date Of Submission:
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Fall 23

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Introduction

There has never been a greater demand for effective and affordable healthcare solutions in a time driven by technological growth and the ongoing pursuit of humanitarian advancement. The Blood Donation Mobile Application, also known as "ULAB Blood Bank," is an essential step in bridging the gap between blood donors and people in urgent need of this life-saving resource. ULAB Blood Bank will be created with the goal of streamlining the blood donation procedure, encouraging voluntary blood donation, and ultimately saving lives. ULAB Blood Bank is ready to have a significant impact on the healthcare industry.

ULAB Blood Bank aims to address the challenges associated with blood donation, including a shortage of donors, inefficient communication channels, and the need for a centralized platform for both donors and recipients. By creating a user-friendly, feature-rich mobile application, ULAB Blood Bank intends to create a user community that empowers individuals to participate actively in blood donation efforts. This application will enable donors to connect with hospitals, and patients in need of blood, fostering a sense of social responsibility and strengthening community support.

SRS

1.1 Purpose

The primary purpose of the ULAB blood bank app is to provide a comprehensive and user-friendly platform for managing and facilitating blood donation and distribution. This software solution aims to address the following key objectives:

- Efficient Blood Donation Management: Enables individuals to register as blood donors, schedule next blood donation date, and maintain a personal donation history.

- Blood Request and Matching: Facilitate the process of requesting blood for medical emergency and match blood donors with recipients in a timely manner.

- User Accessibility: Provide a user-friendly and accessible platform for donors and recipients, enhancing the blood donation experience.

- Notifications and Alerts: Implement a notification system to keep users informed about donation opportunities, emergency requests, and important updates.

1. Community Engagement: Promote blood donation as a community service and encourage more individuals to participate in life-saving blood donation activities.

1.2 Scope

1. Users can log in or create an account to the blood donation app.

2. Registered users can post requests for blood donation.

3. Users should provide essential details such as blood type and mention the emergency level when posting for blood request.

4. Notifications keep users informed and engaged in potential donation opportunities.

5. The app's in built dashboard function displays the number of patients saved through successful blood donations.

6. Users can see a countdown timer to know how many days are left for the next donation.

1.3 **Overview**

The ULAB Blood Bank App is a comprehensive and user-centric software solution designed to address the critical requirements of blood donation, management, and distribution. This application serves as a pivotal platform, connecting blood donors and recipients to create an efficient and responsive ecosystem within the context of blood banking.The following sections provide an overview of the key features and functionalities of the app.

Key Features:

- User friendly design
- User can easily find and choose a donor accordingly by the blood group
- Emergency type (urgent/normal)
- Find with location or address
- Donors information
- Status about the last donation
- See who are available to donate
- Acceptors can contact the donors by phone or chat.

Target Users:

- Donors: Individuals willing to donate blood for the benefit of those in need.
- Recipients: Patients and healthcare professionals seeking blood donations for medical treatment.

2. Overall description:

2.1 Assumptions and dependencies

Assumptions:

1. Internet Connectivity: It is assumed that users of the ULAB Blood Bank application will have access to a reliable internet connection, our application will heavily rely on real-time data exchange and communication between donors, recipients, and healthcare providers.
2. Device Compatibility: We assume that the target users possess smartphones or mobile devices capable of running the application. Devices should meet the minimum hardware and software requirements necessary to support the application's functionality.
3. Privacy and Data Security: It is assumed that users will be concerned about their privacy and the security of their personal and medical data. The application will address these concerns and adhere to applicable data protection regulations.

Dependencies:

1. API Integration: The application will depend on the integration of external APIs to access real-time data, such as communication services for donor and recipient matching, and information about nearby hospitals.
2. Database Management: The application will be dependent on the availability and reliability of the database system. The database will store user profiles, medical history, donation history, and other critical information.
3. User Verification: The application will depend on user verification methods, which may include email or phone number verification, to ensure the authenticity of user-profiles and their willingness to participate in blood donation.
4. User Engagement: The success of the application will depend on user engagement, active participation, and willingness to donate and receive blood. An engaged user base is vital for the application's success.

The application will be developed using Dart as the front-end language and Firebase as the back-end database.

2.2 Apportioning of requirements

Certainly, here's a concise apportioning of requirements for our blood bank app project:

Phase 1: MVP Development

- User Registration and Profile Management
- Next Blood Donation Scheduling
- Blood Request and Matching
- Basic Notifications and Alerts
- Database Structure
- Data Privacy (Basic)
- Compatibility (Mobile)

Phase 2: User Experience Enhancement

- Enhanced Notifications and Alerts
- Usability Enhancements
- Performance Optimization

Phase 3: Data Privacy and Compliance

- Advanced Data Privacy Measures
- Regulatory Compliance (Full implementation)

Phase 4: Community Engagement

- Enhanced Community Engagement Features
- Additional Usability and Accessibility Improvements
- Final Performance Optimization
- Comprehensive System Testing

This phased approach allows for a progressive development and testing cycle, starting with a minimum viable product and incrementally adding features, improvements, and regulatory compliance measures as the project advances.

2.3 External Software Interfaces:

- User Interface
  - We provide a user friendly environment for the user to interact with this system.
  - The GUI will be followed by the industry standard
- Database
  - A database will be created to update and delete data easily.
  - The database system shall meet certain goals to ensure confidentiality, integrity and authenticity
- Error Handling and Logging:
  - The system will provide appropriate error messages to handle errors and have a fail-safe function

3. System requirements

3.1 Interface

- User Interface
- Available Donor Interface
- Available Receipts Interface
- Communication Interface

3.2 Functional requirements:

3.2.1 Functional features

- Create Profile
- Search blood donor as a guest
- Login of donor/acceptors
- Post blood request
- Notifications and Alerts
- Hospital Information
- Can upload previous blood test reports(Not Mandatory)

3.2.2 Non-functional features

1. Performance: The application will have a fast response time, with minimal delays in loading, and handle any sort of process requests. Our system will be designed to handle user loads without noticeably degrading performance. We will also consider how efficiently the system uses resources, such as memory and network bandwidth.
2. Security: To protect the security and privacy of user information, all data transmission, including user profiles and medical records, will be encrypted. To prevent unwanted access to sensitive data and features, we'll put in place strong user authentication and authorization processes. This will guarantee that the data saved on the device is safe and has not been tampered with by unauthorized users.
3. Connectivity and Usability: We will provide limited functionality in offline mode, allowing users to access essential features, such as Dashboard information or previously downloaded data when network connectivity is unavailable. This application will require minimal training for users to understand its functionality and purpose and we will send out clear and helpful error messages to guide users in resolving issues.

4. Other requirements

Mobile App Compatibility:
The app is compatible with mobile devices, providing a convenient platform for users on the go.

These functional features collectively empower the ULAB blood bank app to efficiently manage blood donation processes, improve user engagement, ensure data security, and contribute to the well-being of the community by maintaining a consistent and safe blood supply.

Device Version Requirements:

- Requires Android
  - 9.0 and up
- App permissions
  - Identity
    - read your own contact card
  - Photos/Media/Files
    - read the contents of your USB storage
    - modify or delete the contents of your USB storage
  - Storage
    - read the contents of your USB storage
    - modify or delete the contents of your USB storage
  - Camera
    - take pictures and videos
  - Other
    - receive data from Internet
    - view network connections
    - full network access

5. Estimated Budget

Team	Number of People	Approximate Payment
Design and User Experience	2	80000
Development Team	8	200000
Testing and Quality Assurance	2	100000

6. Benefits

The ULAB blood bank app offers several key benefits, both to users and the broader community:

- Efficient Blood Donation: Simplifies the blood donation process, allowing donors to schedule next donation date and receive reminders, leading to a consistent supply of safe blood.

- Timely Response to Emergencies: The app facilitates quick matching of donors and recipients, ensuring timely blood supply during critical medical situations.

- Community Engagement: Promotes a culture of altruism and community engagement, encouraging more individuals to become blood donors and save lives.

- User Convenience: Provides a user-friendly and accessible platform, making it easy for donors and recipients to manage their roles.

- Mobile Accessibility: Compatibility with mobile devices ensures users can access the platform conveniently from their smartphones, promoting on-the-go engagement.

In summary, the blood bank app serves as a vital tool for blood banks, healthcare institutions, and the public, contributing to a safer and more efficient blood donation and distribution system while fostering community well-being and saving lives.

7. Scalability and extensibility

<u>Scalability</u>:

Implement the necessary methods to design the application's backend which will allow easy expansion of server resources to accommodate a growing user base that might help balance the load to distribute traffic efficiently.

The application could utilize caching techniques to save frequently requested data, eliminating the need to repeatedly ask the server for the same data.

Scalability could be further enhanced by considering microservice architectures, which would make it possible for each component to be scaled separately.

Extensibility:

The application could be built by following a modular structure, where components can be added, modified, or removed with ease. This enables the gradual rollout of new features or upgrades.

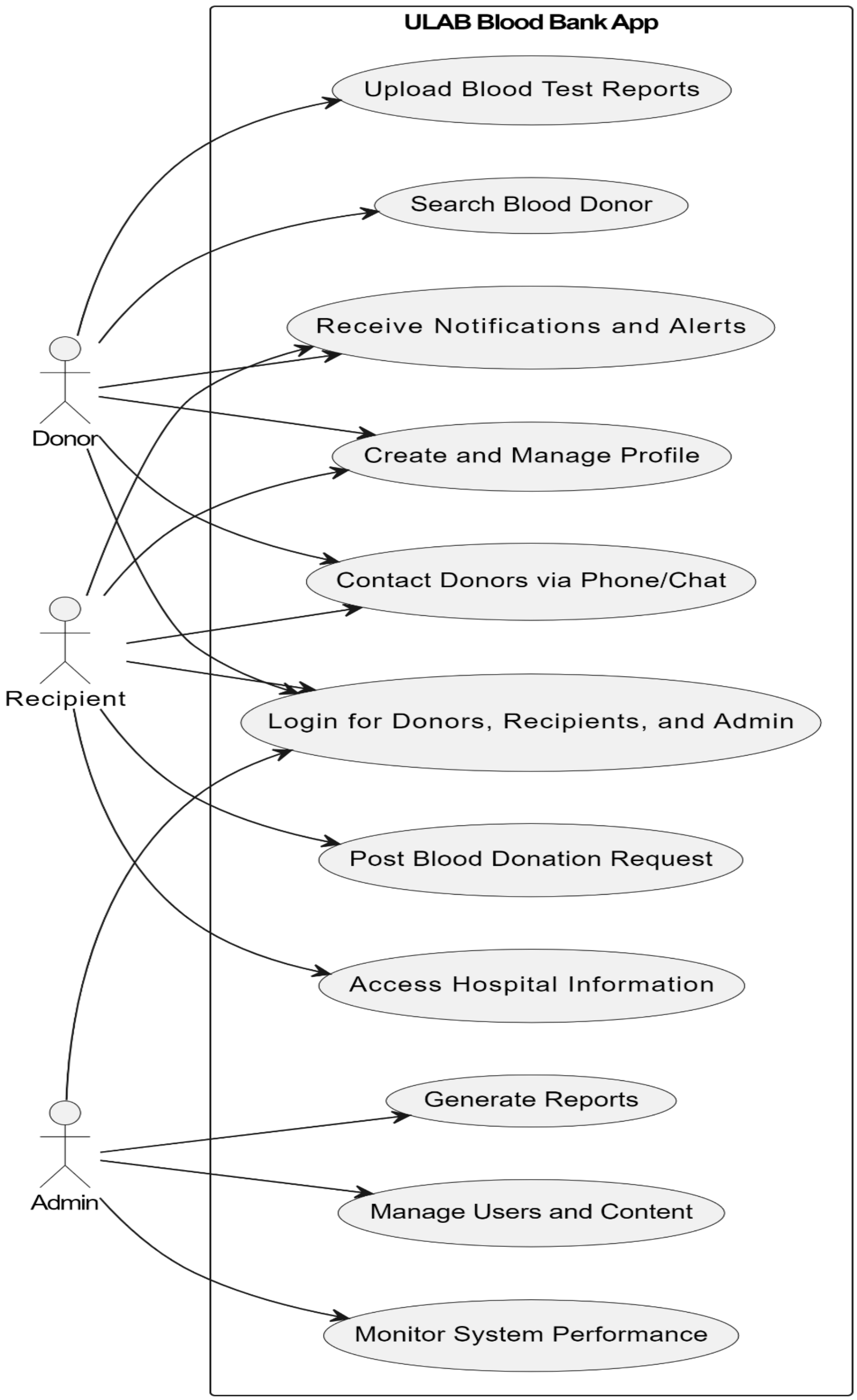
Give external developers access to well-documented APIs and webhooks so they may expand the functionality of the application. This would encourage integrations with external data sources or health monitoring devices.

Developers will be able to integrate a plugin or extension into the system, to build unique plugins or modules to expand the functionality of the program. These plugins must be easy to integrate into the main application.

Provide tools for customization inside the app so users may personalize their experience, such as individualized notification settings, user profiles, or preferences.

The application's informative resources can be increased by allowing users to submit user-generated material, such as articles, tutorials, or community forums.

Use Case


ULAB Blood Bank App
Upload Blood Test Reports
Search Blood Donor
Receive Notifications and Alerts
Create and Manage Profile
Contact Donors via Phone/Chat
Login for Donors, Recipients, and Admin
Post Blood Donation Request
Access Hospital Information
Generate Reports
Manage Users and Content
Monitor System Performance
Donor
Recipient
Admin

Features (Version 1.0)

Note: Which is actually implement before Dec 13, 2023.

- ❖ Onboarding Screen
- ❖ Log in/ Sign Up
    - ➢ Username
    - ➢ Password
    - ➢ Forgot Password
    - ➢ Create Account
- ❖ Countdown timer (days remaining for the next donation)
- ❖ Patients Saved
- ❖ Medical tips / Doctors advice [FAQ: what to take after donation]
- ❖ Who can donate who (table photo)
- ❖ Nearest Hospital Around (includes: Ambulance contacts)
- ❖ Search for blood (posting)
    - ➢ Blood group needed
    - ➢ Urgency type (emergency/normal)
    - ➢ Location
    - ➢ Hospital
    - ➢ Person to contact
- ❖ Profile
    - ➢ Profile Picture
    - ➢ Name
    - ➢ Add Bio
    - ➢ Next Donation (9th sept)
    - ➢ Want to donate (toggle: [status: on/off])
    - ➢ Blood Group
    - ➢ Last Donated (7th july)
- ❖ Contact Info (Under The Profile Section)
    - ➢ Gmail
    - ➢ Contact Number
    - ➢ Address

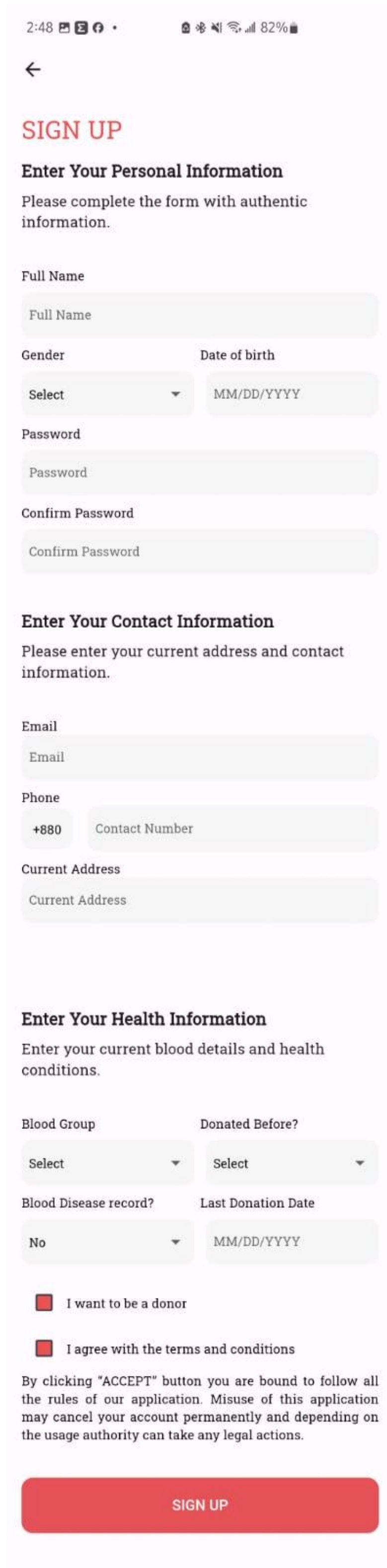
Work Plan

Estimated project development timeline: Sep 28, 2023 To Dec 12, 2023 (Includes weekends and government holidays)

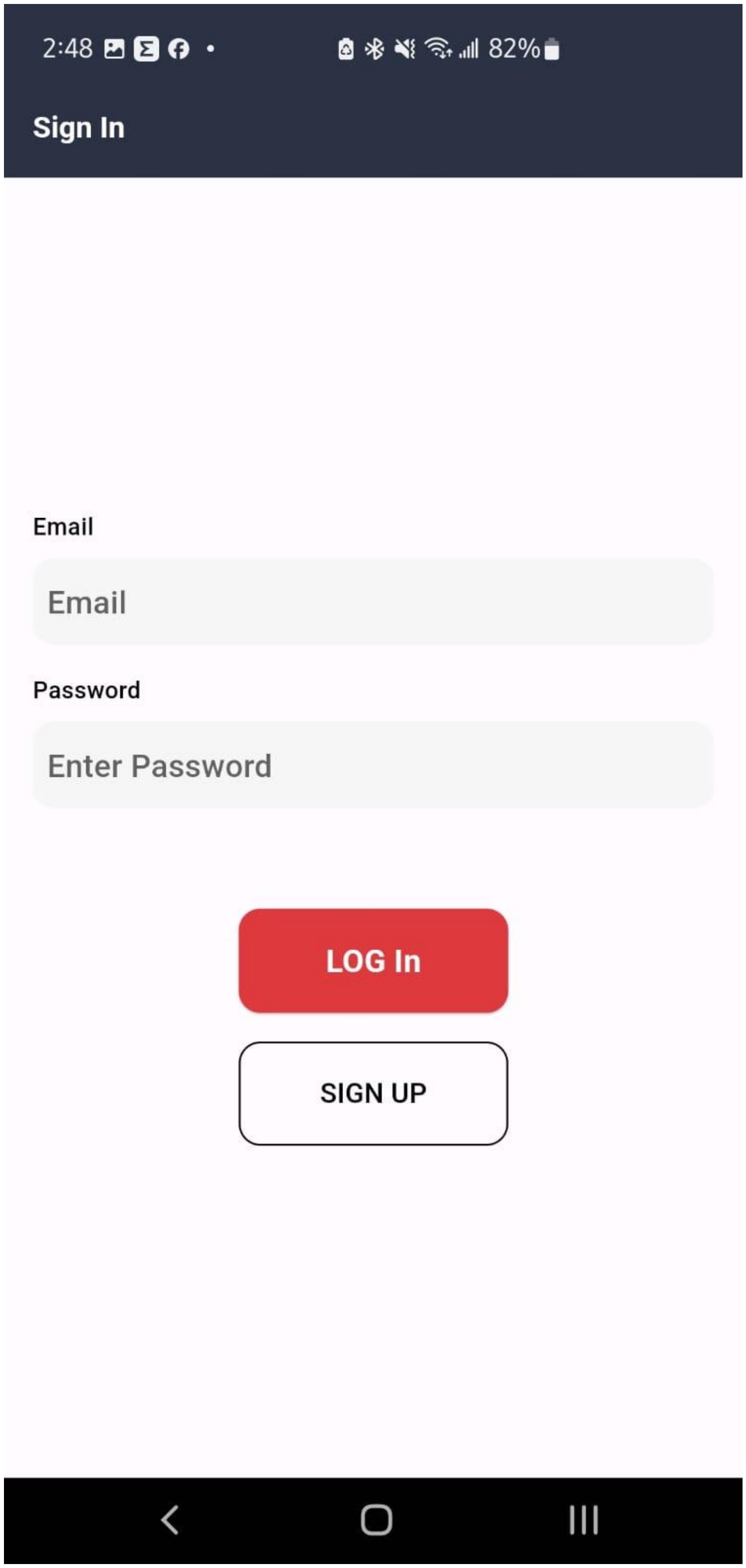
Scope Of Work	Duration	Comment
Idea Presentation	5 Days	**N/A**
SRS UML	7 Days	**N/A**
Outline	14 Days	Prototype using Figma
Feature implementation	46 Days	Programming with flutter to develop the project
Final product submit with documentation	7 Days	Testing and user manual
Total =	**79 Days**	

App Demo

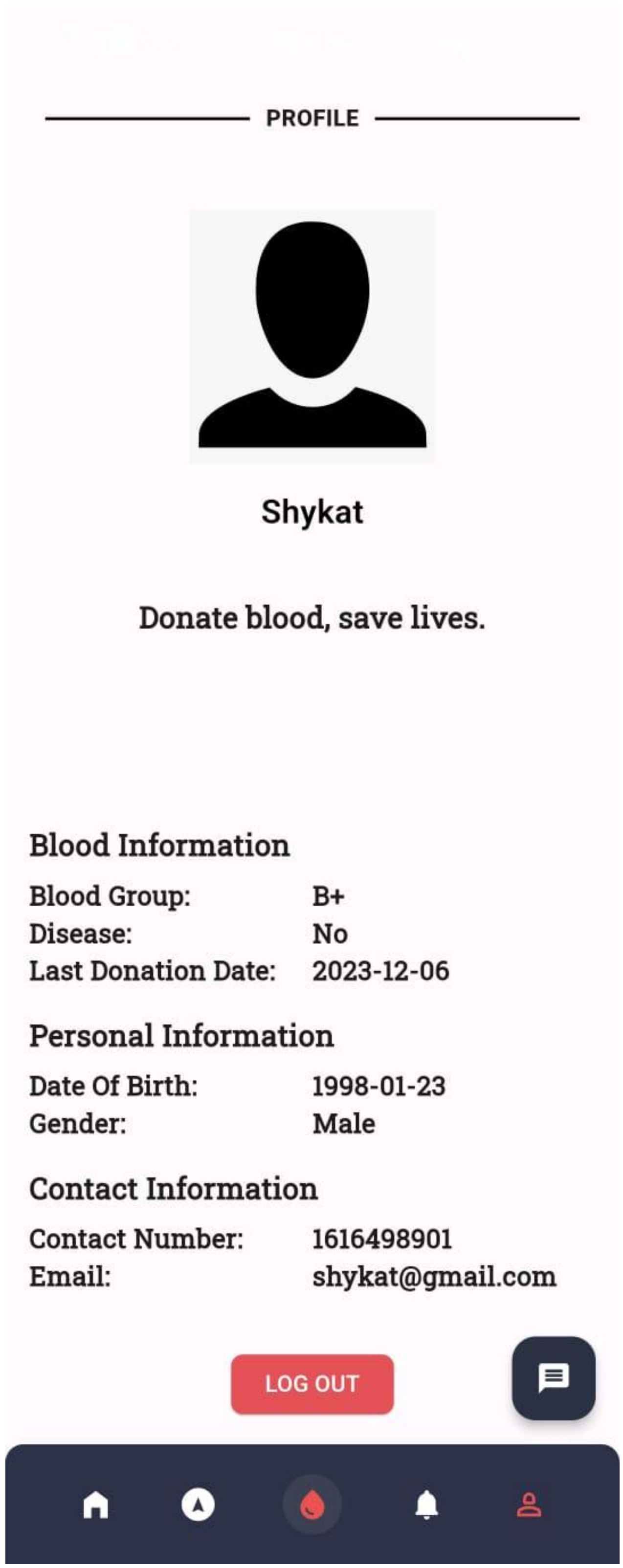
Sign-UP Page:



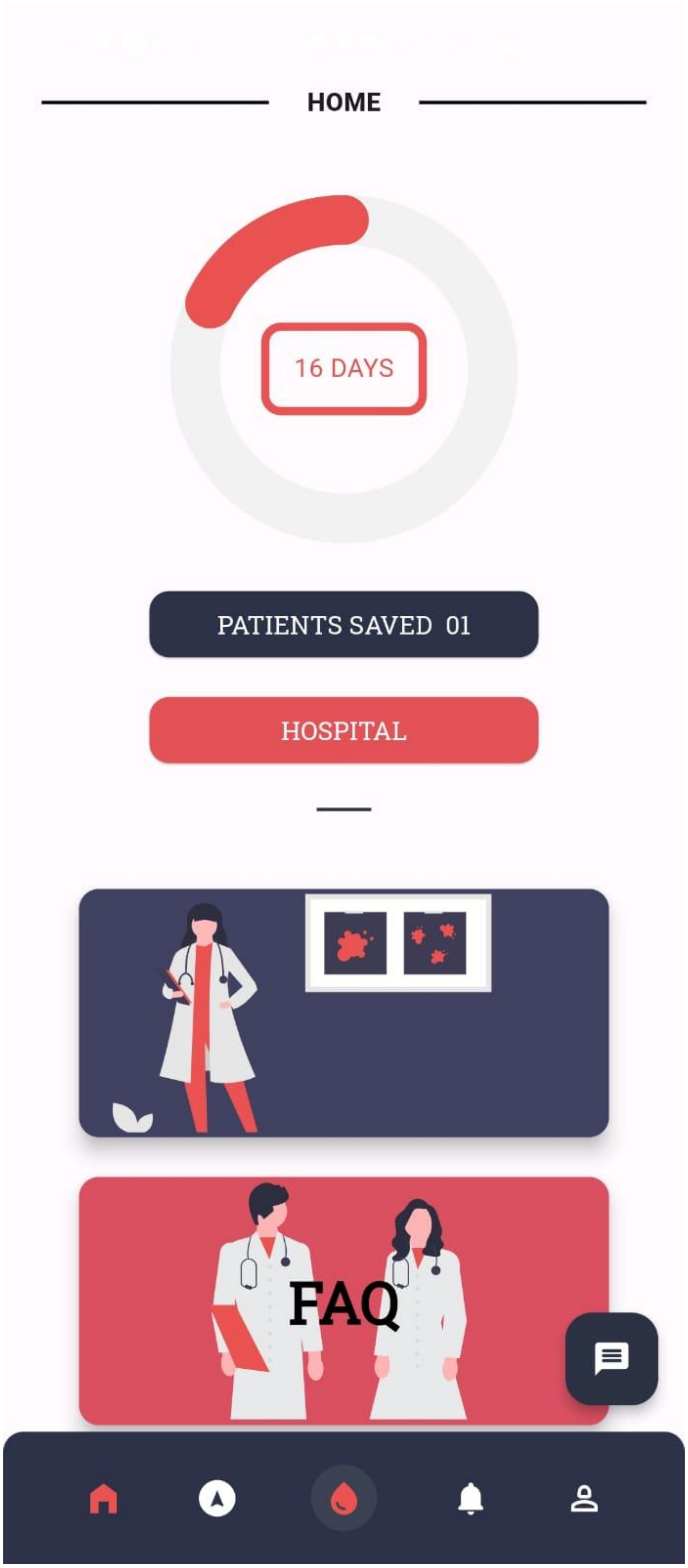
Log In Page:


2:48
82%
Sign In
Email
Email
Password
Enter Password
LOG In
SIGN UP

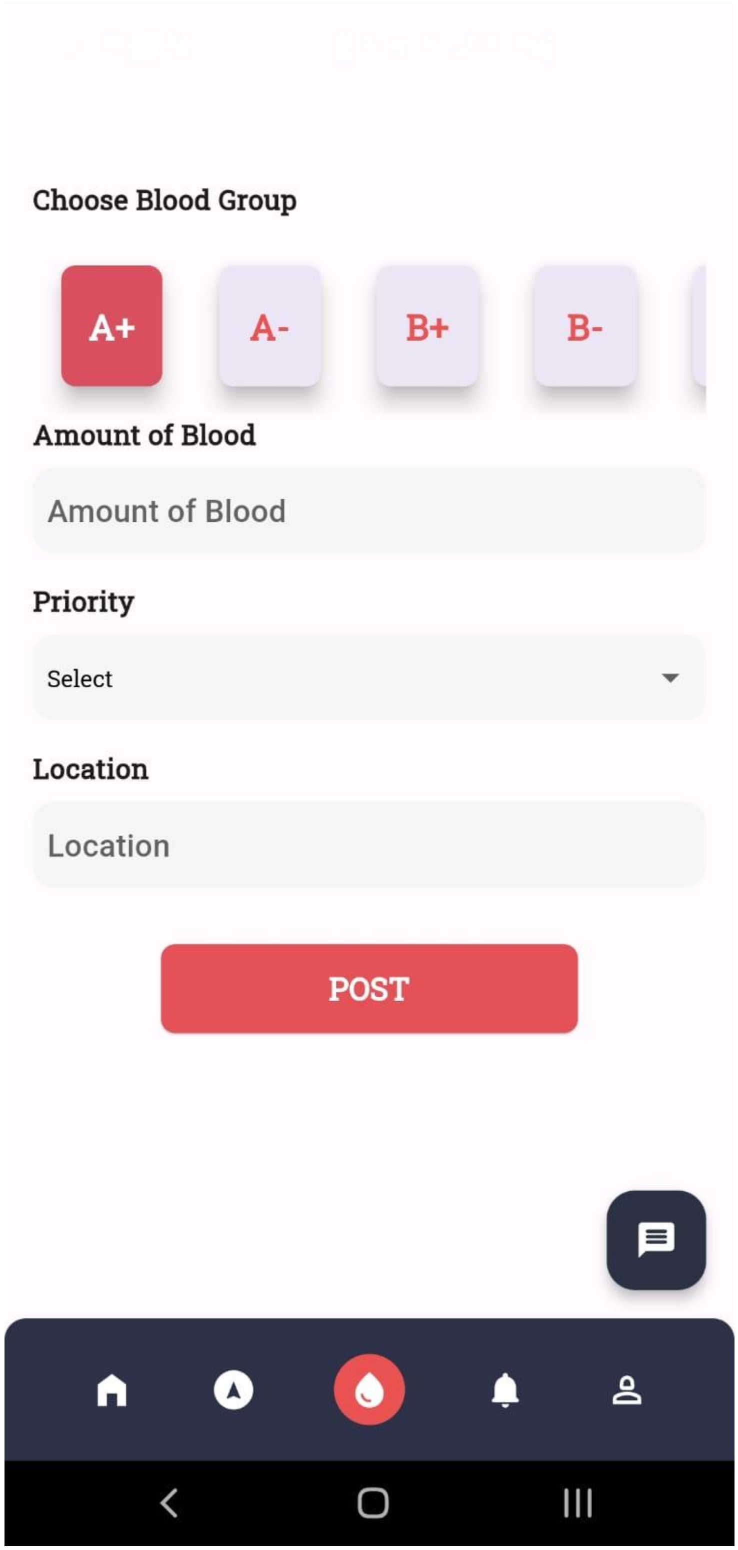
Profile:


PROFILE
Shykat
Donate blood, save lives.
Blood Information
Blood Group: B+
Disease: No
Last Donation Date: 2023-12-06
Personal Information
Date Of Birth: 1998-01-23
Gender: Male
Contact Information
Contact Number: 1616498901
Email: shykat@gmail.com
LOG OUT

Home:

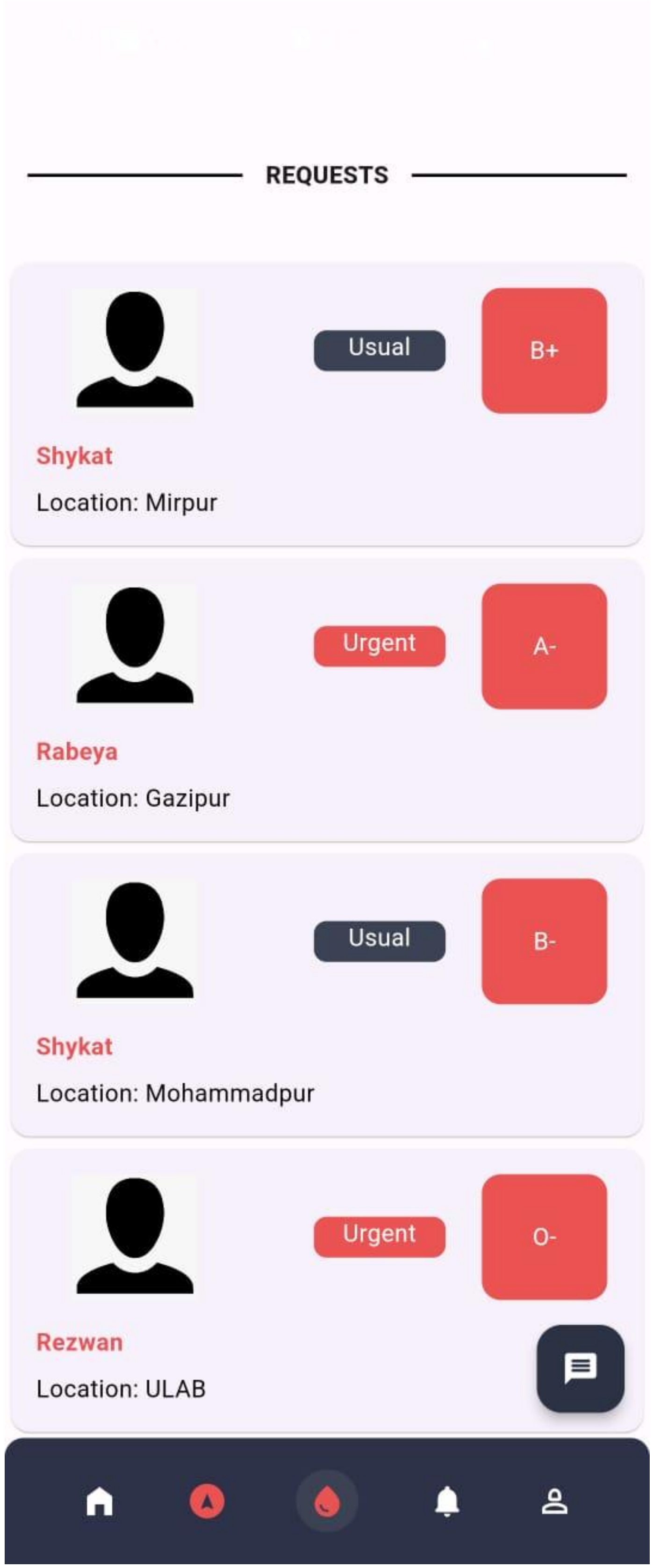


Blood Post:

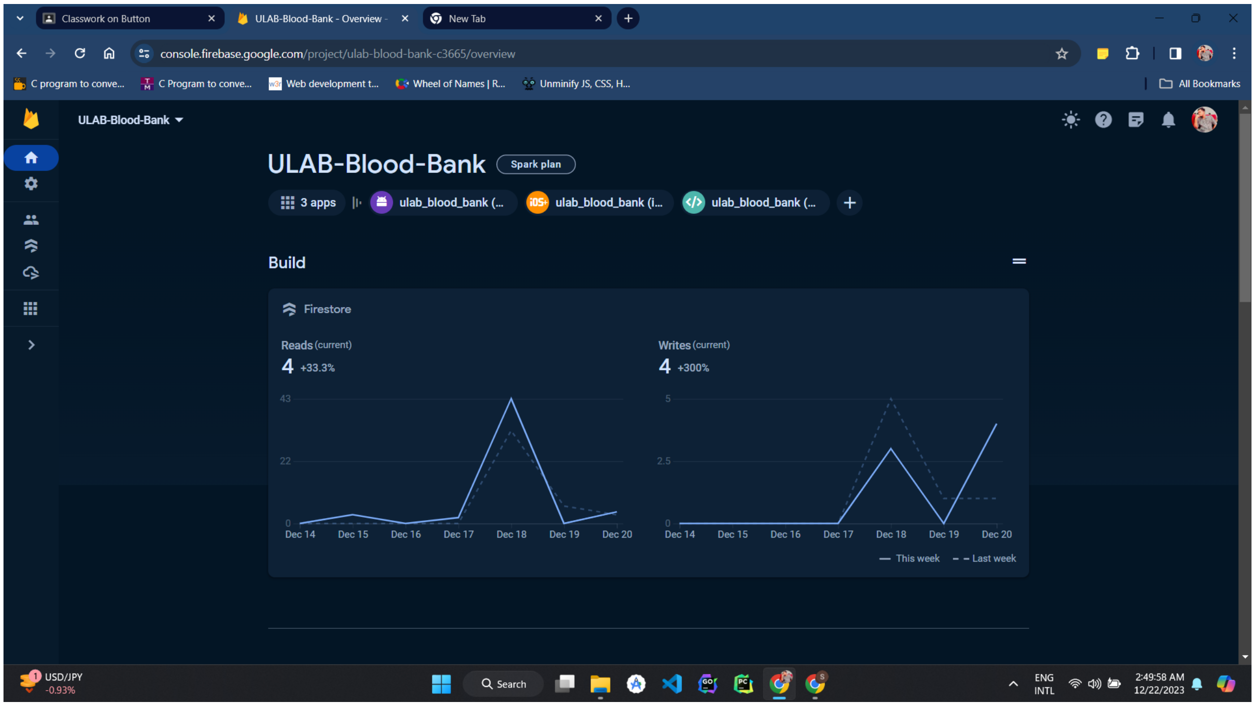


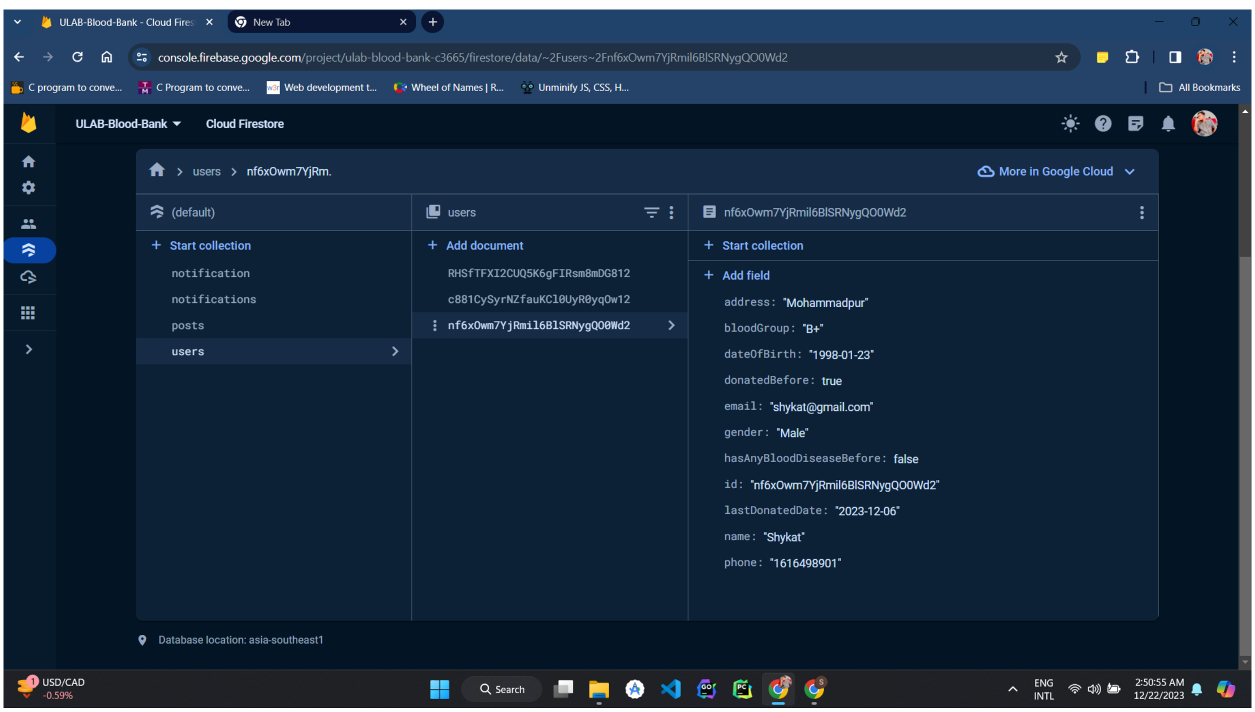
Choose Blood Group
A+
A-
B+
B-
Amount of Blood
Amount of Blood
Priority
Select
Location
Location
POST

Blood Request Page:



Firebase


Classwork on Button
ULAB-Blood-Bank - Overview
New Tab
console.firebase.google.com/project/ulab-blood-bank-c3665/overview
ULAB-Blood-Bank
ULAB-Blood-Bank
Spark plan
3 apps
ulab_blood_bank (...
ulab_blood_bank (i...
ulab_blood_bank (...
Build
Firestore
Reads (current)
4 +33.3%
Writes (current)
4 +300%
Dec 14
Dec 15
Dec 16
Dec 17
Dec 18
Dec 19
Dec 20
This week
Last week


ULAB-Blood-Bank - Cloud Firestore
New Tab
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ULAB-Blood-Bank
Cloud Firestore
users
nf6xOwm7YjRm.
More in Google Cloud
(default)
users
nf6xOwm7YjRmil6BlSRNygQO0Wd2
Start collection
Add document
Start collection
Add field
notification
notifications
posts
users
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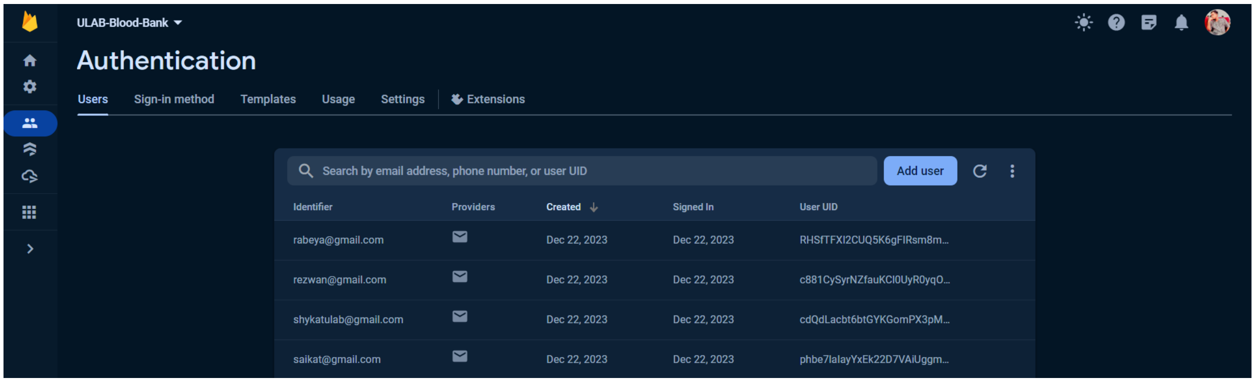
c881CySyrNZfauKCl0UyR0yqOw12
nf6xOwm7YjRmil6BlSRNygQO0Wd2
address: "Mohammadpur"
bloodGroup: "B+"
dateOfBirth: "1998-01-23"
donatedBefore: true
email: "shykat@gmail.com"
gender: "Male"
hasAnyBloodDiseaseBefore: false
id: "nf6xOwm7YjRmil6BlSRNygQO0Wd2"
lastDonatedDate: "2023-12-06"
name: "Shykat"
phone: "1616498901"
Database location: asia-southeast1


ULAB-Blood-Bank
Authentication
Users
Sign-in method
Templates
Usage
Settings
Extensions
Search by email address, phone number, or user UID
Add user
Identifier
Providers
Created
Signed In
User UID
rabeya@gmail.com
Dec 22, 2023
Dec 22, 2023
RHSfTFXI2CUQ5K6gFIRsm8m...
rezwan@gmail.com
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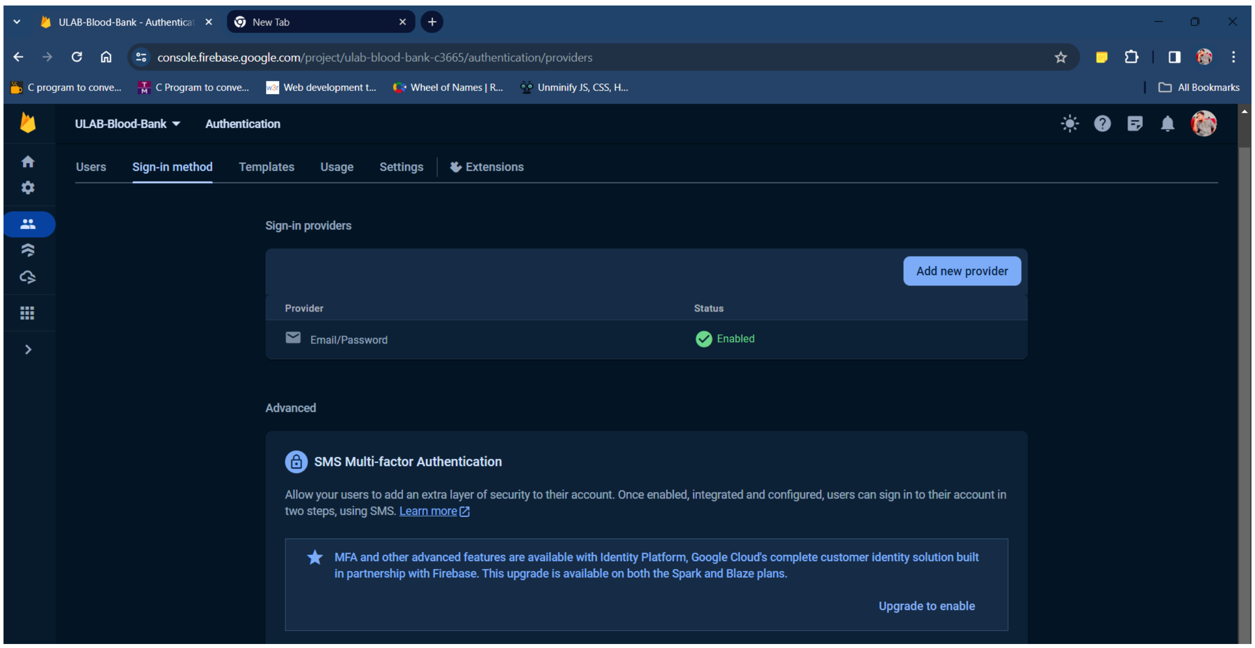
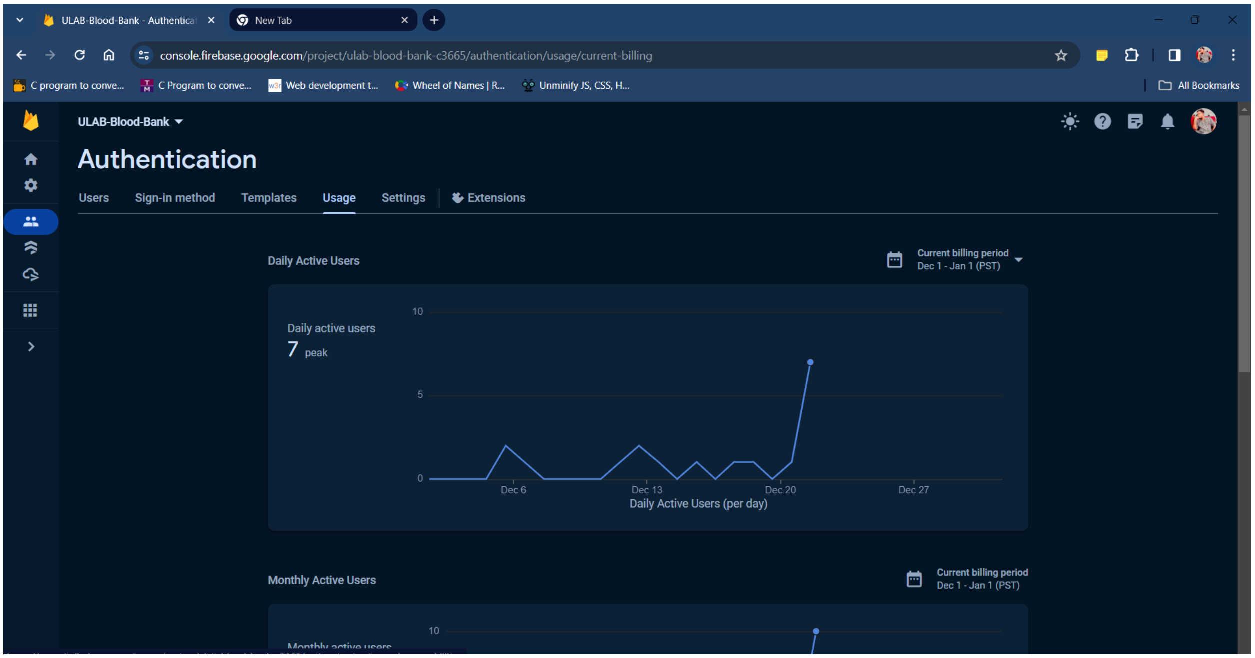

ULAB-Blood-Bank - Authenticat
New Tab
console.firebase.google.com/project/ulab-blood-bank-c3665/authentication/providers
C program to conve...
C Program to conve...
Web development t...
Wheel of Names | R...
Unminify JS, CSS, H...
All Bookmarks
ULAB-Blood-Bank
Authentication
Users
Sign-in method
Templates
Usage
Settings
Extensions
Sign-in providers
Add new provider
Provider
Status
Email/Password
Enabled
Advanced
SMS Multi-factor Authentication
Allow your users to add an extra layer of security to their account. Once enabled, integrated and configured, users can sign in to their account in two steps, using SMS. Learn more
MFA and other advanced features are available with Identity Platform, Google Cloud's complete customer identity solution built in partnership with Firebase. This upgrade is available on both the Spark and Blaze plans.
Upgrade to enable



Conclusion

The "ULAB Blood Bank" mobile application is a comprehensive solution designed to enhance the blood donation process. It aims to bridge the gap between blood donors and recipients by providing a user-friendly, feature-rich platform. This app facilitates efficient blood donation management, timely matching of donors with recipients, and promotes community engagement in blood donation.

The application is structured to be accessible and engaging, encouraging more individuals to participate in blood donation activities.